Global executive summary · SOSistem™

Sistema S.O.S.© · Early warning for human safety · global presentation · version 6.6.3 · October 31, 2026

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Accessibility
Braille · Morse · sign language

OBJECTIVE

To improve the prevention, response and timely identification of emergencies in spaces where no one can see, through sensors and terminals (screens) that instantly share information true to reality, with no cameras and no recording.

WORLD

684,000
deaths per year from falls
WHO, 2021

≈ 3 M
work-related deaths per year
ILO, 2023

40 %
of countries still lack multi-hazard early warning
WMO, 2025

MEXICO

17.7 M
people aged 60 and over
INEGI, 2026

258,522
schools across all levels
SEP, 2025

33,396
earthquakes recorded in 2024
Pacheco, 2025 (SSN)

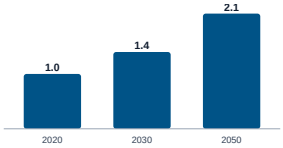
BAJA CALIFORNIA · ENSENADA

3.77 M
inhabitants in Baja California
INEGI, 2021

443,807
inhabitants in the municipality of Ensenada
Secretaría de Economía, n.d.

Pilot
real-world test at ContenedorEns©; proposed for the Port of Ensenada
Ramírez Salgado, 2026

People aged 60 and over worldwide (billions)



Source: WHO (2025).

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Acronyms, symbols and standards

Acronyms

ASIPONA	National Port System Administration (Administración del Sistema Portuario Nacional, Mexico)
ASL	American Sign Language
CAP	Common Alerting Protocol, OASIS standard for emergency alerts
CDC	Centers for Disease Control and Prevention (United States)
CRED	Centre for Research on the Epidemiology of Disasters (UCLouvain, Belgium)
EM-DAT	International disaster database of CRED
ILO	International Labour Organization
IMO	International Maritime Organization
INEGI	National Institute of Statistics and Geography (Instituto Nacional de Estadística y Geografía, Mexico)
ISO	International Organization for Standardization
ITU	International Telecommunication Union
LFPDPPP	Federal Law on the Protection of Personal Data Held by Private Parties (Mexico)
LFT	Federal Labour Law (Ley Federal del Trabajo, Mexico)
LSM	Mexican Sign Language (Lengua de Señas Mexicana)
mmWave	Millimetre-wave radar: detects presence and falls without imaging
NOM	Official Mexican Standard (Norma Oficial Mexicana)
SDG	Sustainable Development Goals of the United Nations 2030 Agenda
SEP	Ministry of Public Education (Secretaría de Educación Pública, Mexico)
SSN	National Seismological Service (Servicio Sismológico Nacional, UNAM, Mexico)
UMA	Unit of Measure and Update (Unidad de Medida y Actualización, Mexico)
UN	United Nations
UNDRR	United Nations Office for Disaster Risk Reduction
WHO	World Health Organization
WMO	World Meteorological Organization

Symbols and criteria for figures

≈	Approximate or rounded value; not an exact measurement.
M	Millions (10^6).
B	Billions (10^9).
%	Percentage of the total reported by the source.
×	Times: ratio between two values.
≤	Equal to or less than the stated time.
N0–N3	System alert levels: N0 on site, N1 person in charge, N2 monitoring centre, N3 911.
Own calculation	Result obtained by the author from the cited sources; indicated next to the figure.

Numbers and symbols follow the international standard on quantities and units ISO 80000-1 and, in Mexico, NOM-008-SCFI-2002 (General System of Units of Measurement). Data come from official sources or international organizations and are cited by author and year.

ISO 80000-1:2022 · NOM-008-SCFI-2002

1. Executive summary

SOSistem™ detects within seconds a person on the floor, an earthquake, a gas leak or a violent situation in restrooms, changing rooms, cabins and confined spaces, with no cameras and no recording, and escalates the alert all the way to 911.

The problem

Every year **684,000 people die from falls** and **37.3 million** falls require medical attention (WHO, 2021). Nearly **3 million people** die from work-related causes (ILO, 2023). Where law and dignity rule out cameras, an emergency can go unnoticed for critical minutes.

The solution

Imageless sensors (mmWave radar, gas, smoke, oxygen, sound and vibration) and terminals that report instantly. The alert escalates by level to the person in charge, the monitoring centre and **911** with a CAP message, and the equipment runs for up to 30 days without grid power.

The scale

82 % of people aged 10 and over own a mobile phone (ITU, 2025): every phone can be a sensor, a screen or both. In Mexico alone there are **258,522 schools** (SEP, 2025) and **17.7 million** people aged 60 and over (INEGI, 2026).

The alignment

Contributes to SDGs 3, 8.8 and 11.5, to target (g) of the Sendai Framework and to the “Early Warnings for All” initiative (see chapter 7). Where early warning is limited, disaster mortality is **nearly six times higher** (UNDRR and WMO, 2024).

What sets SOSistem™ apart

No cameras	Only states and measurements; no images, video or voice recording.
Real time	Tiered N0–N3 alert; immediate upon a confirmed fall or an earthquake.
For everyone	Light, sound, 22 languages, braille, Morse and sign language.
Resilient	Battery of up to 30 days; works with or without a network.
Scalable	From one person's phone to an entire building, a city or a port.

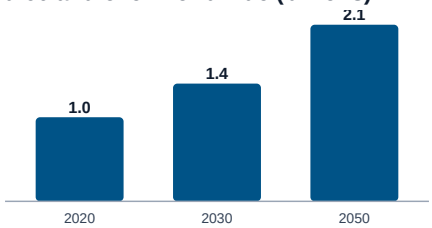
CURRENT STAGE

Working prototype and public test Terminal at sosistem.com/terminal.html. Physical pilot at the ContenedorEns© offices (Ensenada, Baja California, Mexico).

OMS/WHO (2021) · OIT/ILO (2023) · UIT/ITU (2025) · SEP (2025) · INEGI (2026) · UNDRR & OMM/WMO (2024) · OASIS (2010)

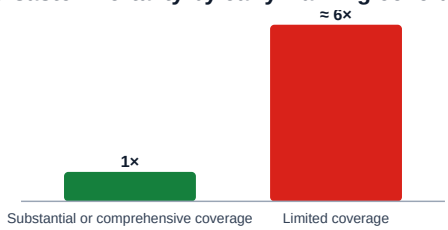
2. The global challenge in figures

People aged 60 and over worldwide (billions)



Source: WHO (2025).

Relative disaster mortality by early warning coverage



Source: UNDRR and WMO (2024): nearly six times higher where early warning is limited.

People and ageing

8.2 B
people on the planet in 2024
United Nations, 2024

≈ 853 M
people aged 65 and over (10.4 % of the population)
Own calculation with World Bank, 2025

1.4 B
people aged 60 and over in 2030: 1 in 6
WHO, 2025

426 M
people aged 80 and over in 2050
WHO, 2025

14 M
adults aged 65+ who fall each year in the United States alone (1 in 4)
CDC, 2026

82 %
of people aged 10 and over own a mobile phone
ITU, 2025

Falls and work

684,000
deaths per year from falls; second leading cause of unintentional injury death
WHO, 2021

37.3 M
falls per year requiring medical attention
WHO, 2021

≈ 3 M
deaths per year from occupational accidents and diseases; 330,000 from accidents
ILO, 2023

395 M
workers with non-fatal injuries per year
ILO, 2023

$586,550
MXN, maximum fine for violating occupational safety and health standards in Mexico (5,000 UMA)
LFT, art. 994; INEGI, 2026

Disasters and early warning

393
disasters recorded in 2024
CRED, 2025

16,753
disaster deaths in 2024
CRED, 2025

167 M
people affected and USD 242 B in losses (2024)
CRED, 2025

119
countries with multi-hazard early warning (60 %)
WMO, 2025

Mexico

17.7 M
people aged 60 and over in Mexico
INEGI, 2026

258,522
schools; 32.5 M students and 2.06 M teachers
SEP, 2025

33,396
earthquakes recorded in Mexico in 2024
Pacheco, 2025 (SSN)

117
seaports in Mexico
The Logistics World, 2024

UN (2024) · World Bank (2025) · OMS/WHO (2021, 2025) · CDC (2026) · OIT/ILO (2023) · LFT art. 994 · INEGI (2026) · UIT/ITU (2025) · CRED (2025) · OMM/WMO (2025) · UNDRR & OMM/WMO (2024) · SEP (2025) · Pacheco (2025) · The Logistics World (2024)

3. Where it can operate: scope of service

SOSistem™ works wherever people are in spaces that cannot be monitored with cameras. The table presents the **scope of service** with official figures and an **illustrative 1 % scenario** to gauge reach. It is not a sales projection: scope and investment are defined according to the needs of each site.

Universe	Size	Source	1 % scenario
Schools in Mexico (all levels)	258,522	SEP, 2025	2,585 schools
Students and teachers in Mexico	34,577,132	SEP, 2025	≈ 345,771 people
Schools mapped worldwide	> 2.2 M	Giga, n.d. (140 countries)	22,000 schools
People aged 60+ in Mexico	17,700,000	INEGI, 2026	177,000 people
People aged 65+ worldwide	≈ 853 M	World Bank, 2025 (own calculation)	≈ 9 M people
Inhabitants of Ensenada, B.C.	443,807	Secretaría de Economía, n.d.	≈ 4,438 people
Seaports in Mexico	117	The Logistics World, 2024	Pilot: Port of Ensenada
National governments	193	UN Member States	2 countries

Priority sectors

Schools and day-care centres · Hospitals and medical tourism · Care homes and residences for older persons · Government offices serving the public · Ports, customs and bonded facilities · Industry and maquiladoras · Restaurants and hotels · Passenger and cruise terminals · Stadiums, places of worship and event venues · Shelters and temporary refuges · Warehouses and confined spaces · Homes: every person's mobile phone

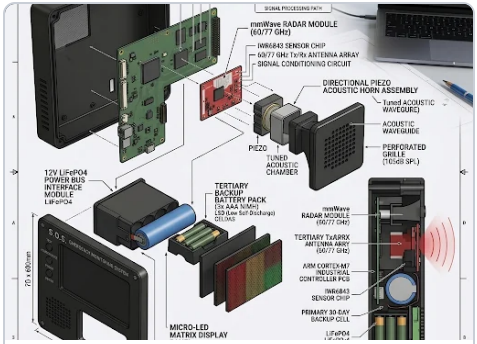
Scope, investment and service are defined according to the needs of each site, following a free assessment. The financial proposal is presented separately.

SEP (2025) · Giga (s. f./n.d.) · INEGI (2026) · World Bank (2025) · Secretaría de Economía (s. f./n.d.) · The Logistics World (2024) · UN (2024)

4. The solution: sensors, terminals and screens



NODO S.O.S.© terminal: screen, mmWave radar and siren



Components and 30-day power backup



S.S.O.S.© screen at the restroom entrance



Person on the floor



Earthquake: evacuation route



People count



Gas leak

How it alerts

Level	Time	Alert
N0	2 min	On site: light, sound and voice
N1	4 min	Person in charge of the site
N2	7 min	Monitoring centre
N3	10 min	911 with CAP message

Upon a confirmed fall or an earthquake, the alert is immediate. Safety colours in accordance with ISO 7010, ISO 3864-1 and NOM-026-STPS-2008.

Twelve services

1. People count
2. Distribution and dwell time
3. Person on the floor
4. Maximum occupancy exceeded
5. Zones in m² and m³
6. Executive screens
7. Earthquake, explosion and calls for help
8. Gases and oxygen
9. 911 link and event log
10. Executive statistics
11. Disaster and committee
12. Light, sound and public address

Images: renders and reference screens of the system. ISO 7010:2019 · ISO 3864-1:2011 · NOM-026-STPS-2008 · OASIS CAP 1.2

5. Every phone a sensor: family help around the clock

Mobile phones, tablets, computers and other devices, online or offline, can work as **sensors**, as **observation and information terminals (screens)**, or as **both at once**.

IMPACT

Anyone can download SOSistem™ to their phone, activate it and link it to a family member 24 hours a day, 7 days a week: upon a fall, upon the words “S.O.S.©” or “help”, or when the system itself detects the fall, the family member is alerted instantly. With 82 % of people aged 10 and over owning a mobile phone (ITU, 2025), protection can reach billions of people without installing a single device.

No video, voice or conversations are recorded: it is only real-time help.

How it works

1. **Download and activate** SOSistem™, no sign-up required.
2. **Link with a family member** using the same phrase on both devices, only once.
3. **Detect** the fall, the words for help or immobility, even if the person cannot touch the screen.
4. **Alert instantly** the family member; if no one responds, the alert escalates to 911.
5. **Cancel by voice**: “all good” stops a false alarm.

Sensors · current options

On the phone	Accelerometer (fall and impact), gyroscope (rotation and tilt), microphone (recognizes only “S.O.S.©”, “help”, “it's shaking” and “all good”; it does not record), location with permission, immobility and pocket mode.
Fixed	mmWave radar, motion, distance, sound level, smoke, gas, oxygen, hydrogen, ammonia, heat, cold, light and darkness, water on the floor, mass, space and capacity, people count and dwell times.

Status: the phone Terminal is available as a public test version, with voice always active and phone-to-screen linking by phrase; the permanent 24/7 family link is under development.

UIT/ITU (2025) · LFPDPPP (2025)

6. Accessibility: 22 languages, braille, Morse and sign language

In an emergency, not everyone can hear, see or speak. SOSistem™ delivers every alert through four complementary channels so that alerts and help reach everyone.

22 languages	Alerts and screens in each person's language: UN official languages plus European, Asian and regional languages.
Braille	Tactile signage and a way to call for help without seeing the screen.
Morse	Universal distress signal (SOS) in light and sound, in line with the ITU.
Sign language	Alerts for deaf people: Mexican Sign Language (LSM); American Sign Language (ASL) in development.

Braille: how to call for help without seeing the screen (proposal)

- **Braille plate** next to each terminal and in the cabins, with the word “SOS” and directions to the help button.
- **Raised tactile button**: holding it down for 3 seconds sends the alert, even if the person cannot see or speak.
- **Vibration and sound confirmation**: a pattern felt in the hand indicates that help has been alerted; useful for blind or deafblind people.
- **On the phone**: alert with a gesture (triple tap) and reading via VoiceOver or TalkBack; the user's braille display shows the alert status.

Proposal under development; it will be validated with blind, deaf and deafblind people and their organizations.

ITU-R M.1677-1 (2009)

7. Impact, global alignment and indicators

Global framework	What it says (official text)	SOSistem™ contribution
SDG 3 · Good health and well-being	*“Ensure healthy lives and promote well-being for all at all ages.”*	Reduces the time between a fall or medical crisis and care.
SDG 8, target 8.8 · Decent work	*“Protect labour rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment.”*	Monitors confined spaces, gases and occupancy in workplaces.
SDG 11, target 11.5 · Resilient cities	*“By 2030, significantly reduce the number of deaths and the number of people affected […] caused by disasters, including water-related disasters, with a focus on protecting the poor and people in vulnerable situations.”*	Earthquake alerts, evacuation routes and people counts per facility.
Sendai Framework, target (g)	*“Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information and assessments to people by 2030.”*	Brings early warning inside buildings.
Early Warnings for All (UN-WMO)	*Everyone on Earth protected by early warning systems by 2027.*	Takes the alert to the “last metre”: the cabin, the restroom and each person's phone.

Outcome indicators

Indicator	Measurement
Alert to the person in charge	≤ 4 min; immediate for a fall or earthquake
Alert to 911 (CAP)	≤ 10 min
People counted per facility	In real time, without identifying them
False alarms	Percentage per event, with an auditable log
Facilities and people protected	Count by country, sector and period

SWOT analysis

Strengths Sees without cameras · alert in seconds · 30 days off-grid · phone as sensor	**Weaknesses** Pilot stage · conformity assessment under way · initial equipment cost
Opportunities Early Warnings for All · population ageing · legal duty to prevent	**Threats** Unnoticed emergencies · earthquakes without warning · false alarms (mitigated by levels)

Privacy and standards

No cameras or recording; only the real-time alert is transmitted (event type, time and, if authorized, location), in accordance with the LFPDPPP (2025). Reference standards: ISO 7010:2019, ISO 3864-1:2011, NOM-026-STPS-2008, ISO 8201, ISO 7731 and OASIS CAP 1.2.

UN (2015) · UNDRR (2015) · UNDRR & OMM/WMO (2024) · OMM/WMO (2025) · LFPDPPP (2025) · ISO (2011, 2019) · STPS (2008) · OASIS (2010)

8. Roadmap and partnerships

Phase	Scope	Outcome
1 · Pilot	Prototype and real-world test at ContenedorEns© (Ensenada, B.C., Mexico).	Alert times and false alarms measured in real operation.
2 · Port	Additional test at the Port of Ensenada, subject to authorization by ASIPONA Ensenada.	Validation in port and bonded facilities.
3 · Conformity	Conformity assessment before national and international bodies.	Equipment ready for installation in regulated sectors.
4 · Scale	Schools, hospitals, government and mobile phones in Mexico and Latin America.	Country-by-country replication with institutional partners.

Investment and service

Scope, investment and service are defined according to the needs of each site, following a free assessment. The financial proposal is presented separately.

SOSistem™ does not replace official emergency services.

Call for partners

- **Governments and civil protection:** pilots in schools and public-facing offices.
- **Development banks and funds:** equipment financing for pilots and conformity assessment.
- **Multilateral organizations:** regional replication within “Early Warnings for All”.
- **Academia:** scientific validation of alert times and false alarms.

Víctor Hugo Ramírez Salgado
Author of Sistema S.O.S.© and SOSistem™

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Ramírez Salgado (2026)

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